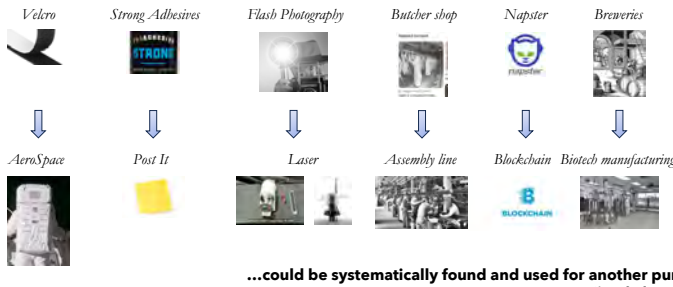


In fact, most "innovations" start out using something invented for something else

Combinatorial Innovation: What if technology wasn't *the product* but the problem-solving parts?

What if technologies conceived/discovered when looking for a specific something...



What if every technology...



Could be seen as a set of parts



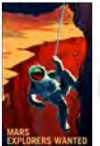
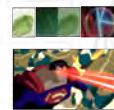
ready to be recombined into a new purpose?



AN EXAMPLE OF A TECH SHELF iTEAMS TECHNOLOGIES REPURPOSED 2020-2024

We've set up "Innovating Factories" over 40 times.

A vision, a Technology + Cross-discipline Team + Robust Opportunity Based on fact Solving a problem + Scale up Blueprint + "Multi-lingual" Innovators, Explorers of Impact



iTEAMS - Visual Syllabus



2008-2018 Fact-sheet

200+ Research breakthroughs
150+ Labs @MIT and Beyond
1000+ alumni from MIT's 5 schools
(and beyond: Harvard, BU, tufts, Navy...)



20+% of projects spun out:

FROM all disciplines

including biotech, nanomaterials, photonics, robotics, nuclear engineering, civil engineering, artificial intelligence, architecture, and many others.

TO all kinds of industries

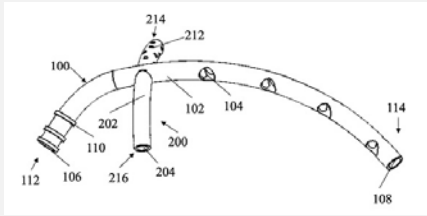
including healthcare, manufacturing, telecommunications, gaming, chemical synthesis, mining, biochemical manufacturing, ...

Many alumni use their newly acquired skills to find technologies to solve new problems



Mechanical Engineering

Minimally Invasive Modular Glaucoma Implant



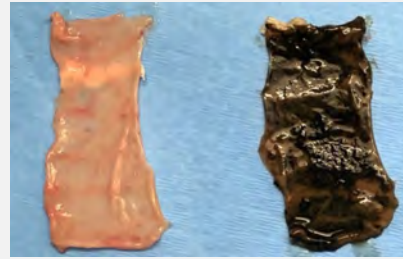
PIs: Matthew Cavuto

Departments:
Mechanical
Engineering/
Imperial College

Link:
<https://patentimage.s.storage.googleapis.com/5d/16/14/1e15fa8947f6af/US20210137736A1.pdf>



Gastrointestinal Synthetic Epithelial Linings

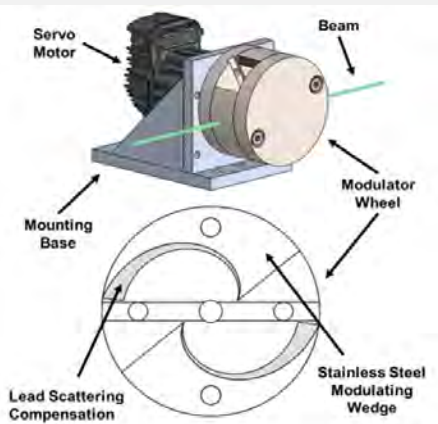


PIs: Gio Traverso, Junwei Li

Department:
Mechanical
Engineering

Links:
<https://www.nih.gov/news-events/nih-research-matters/oral-solution-forms-synthetic-gi-lining>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8221077/>

Proton Beam Energy Modulator

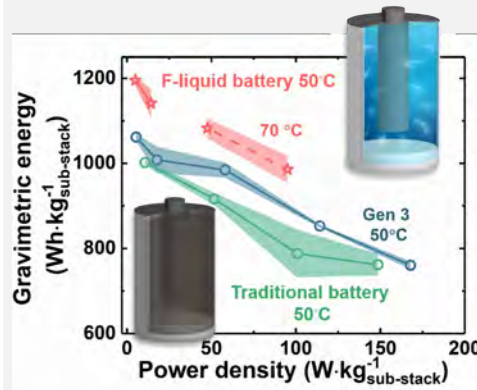


PIs: Matthew Cavuto

Departments:
Mechanical
Engineering/
Imperial College

Link:
<https://patentimage.s.storage.googleapis.com/6c/af/76/79e2071fb053e/US1141608.pdf>

F-Liquid Cathode to Boost Primary Battery Energy Density

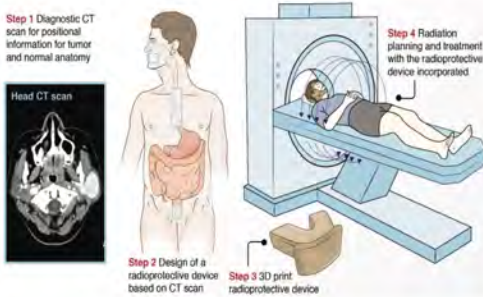


PIs: Betar M Gallant, Haining Gao (Lab Liaison)

Departments:
Mechanical
Engineering/
Material Science
and Engineering

Links:
<https://pubs.acs.org/doi/10.1021/acs.jpcc.8b00569>
<https://onlinelibrary.wiley.com/doi/10.1002/aenm.201900393>

Personalized Radiation Attenuating Devices for Gastrointestinal Mucosal Protection

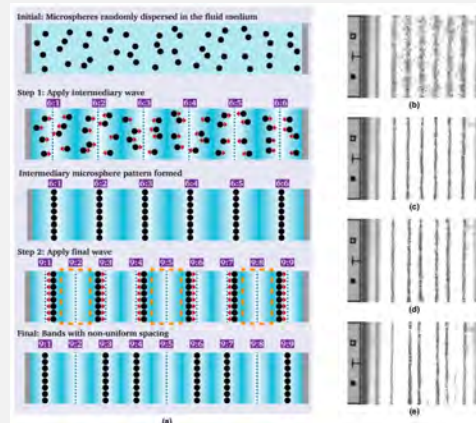


PIs: James Byrne (UOI), Giovanni Traverso (MIT)

Departments:
Mechanical
Engineering/
University of Iowa

Link:
<https://doi.org/10.1080/17434440.2018.1510771>

Acoustophoresis



PI: Brian Anthony

Department:
Mechanical
Engineering

Link:
<https://asmedigitalcollection.asme.org/IMECE/proceedings-abstract/IMECE2017/58356/V002T02A074/265680>

Mechanical Engineering

Extending the Life of Low-Cost, Compact, Lightweight Batteries



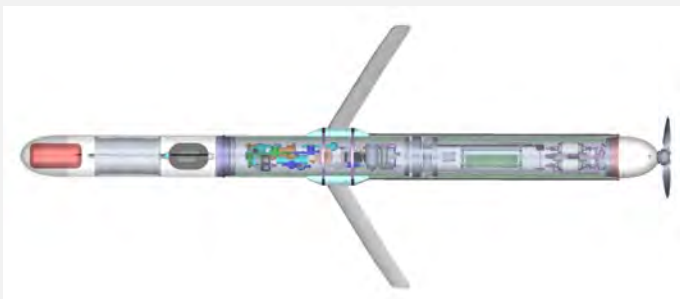
PIs: Yang Shao-Horn

Department: Mechanical Engineering

Links:

<https://meche.mit.edu/news-media/extending-life-low-cost-compact-lightweight-batteries>

Autonomous Hydrogen Powered Underwater Vehicles for the Blue Economy



PI: Douglas Hart

Departments:

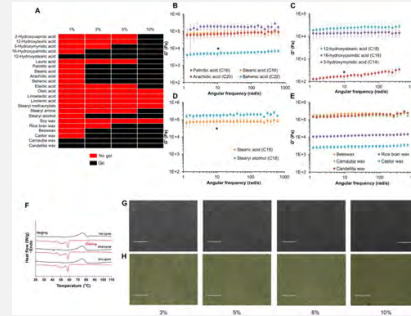
Mechanical Engineering; Beaverworks

Links:

https://eospso.nasa.gov/sites/default/files/publications/UnderstandingEarth-OurOcean_508_0.pdf

<https://www.media.mit.edu/projects/my-deepsea-mybackyard/overview/>

Molecular Gastronomy Inspired Drug Delivery Systems



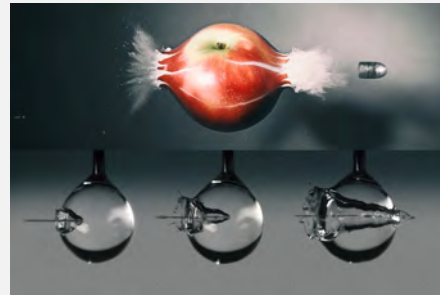
PIs: Gio Traverso, Bob Langer, Ameya Kirtane

Departments:
Mechanical Engineering,
Chemical Engineering, Brigham and Women's Hospital

Link:

<https://doi.org/10.1126/sciadv.abm8478>

Small Jet Injector



PIs: David Fernandez Rivas

Departments:
Mechanical Engineering /
Utwente

Link:

<https://pubs.aip.org/aip/jap/article-abstract/127/10/104901/281690/Microfluidics-control-the-ballistic-energy-of?redirectedFrom=fulltext>

Low-Cost Transfemoral Rotator



PIs: Matthew Cavuto

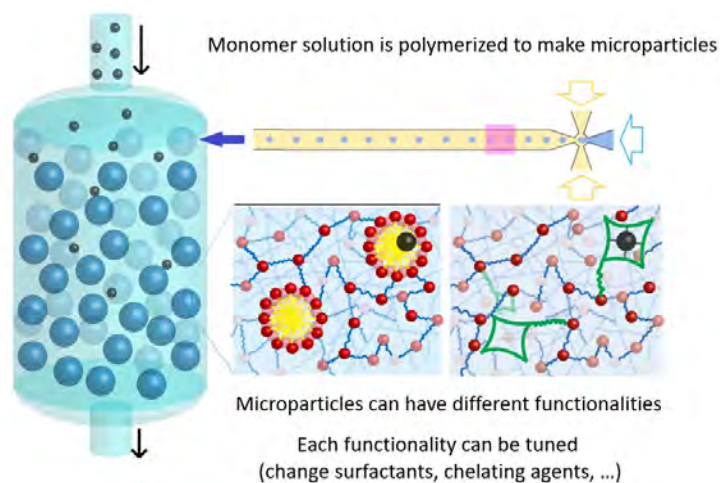
Departments:
Mechanical Engineering /
Imperial College

Links:

<https://patentimages.storage.googleapis.com/f8/af/e3/a5296982babeddb/US10639171.pdf>

Chemical Engineering

Sustainable Hydrogels For Water Treatment



Pl: Pat Doyle

Department: Chemical Engineering

Links:

<https://news.mit.edu/2022/j-wafs-150k-solutions->

[grant-rapid-removal-micropollutants-0907](#)

<https://news.mit.edu/2024/zwitterionic-hydrogels->

[swiftly-eliminate-micropollutants-from-water-0109](#)

<https://pubs.rsc.org/en/content/articlelanding/2024/s>

[u/d4su00052h](#)

Synthetic Approach to Porous Carbon Electrodes for Specific Electrochemical Applications

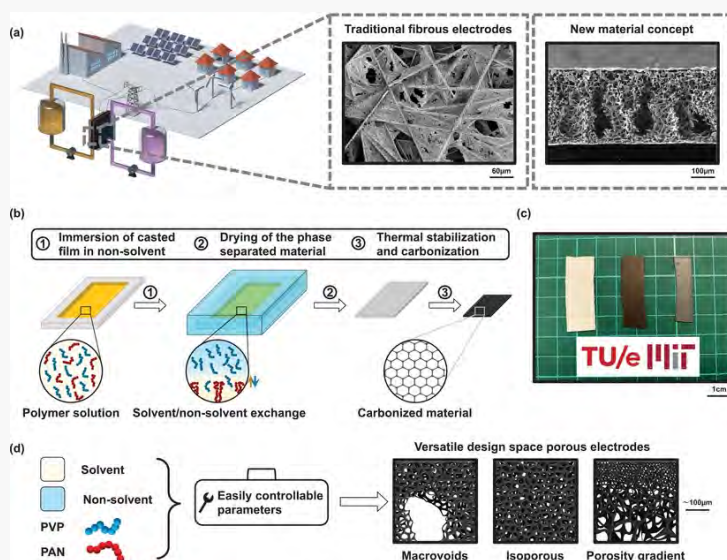


Image retrieved from:

<https://www.researchgate.net/publication/349736284/figure/fig1/AS:11431281173268680@1688822373282/Concept-for-NIPS-fabricated-porous-electrodes-integrated-into-a-RFB-a-RFB-powered-by.png>

PI: Fikile R.

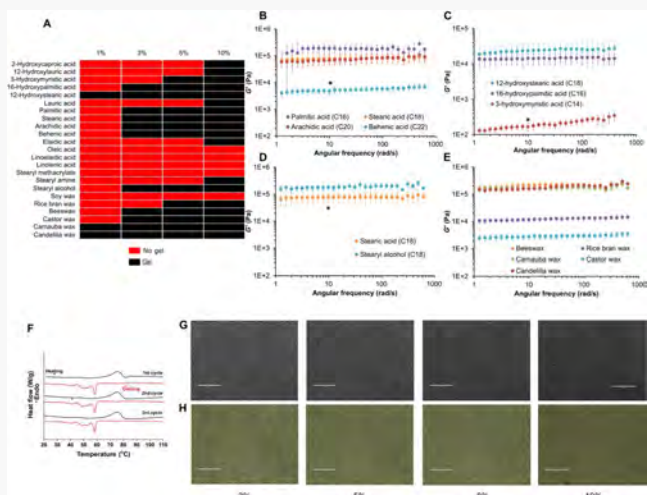
Department:

Chemical Engineering

Link: [https://www.sciencedirect-](https://www.sciencedirect.com/science/article/pii/S0950268820300091)

com.libproxy.mit.edu/science/article/abs/pii/S2451910319301668

Molecular Gastronomy Inspired Drug Delivery Systems



Pls: Gio Traverso, Bob Langer, Amaye Kirtane

Department: Chemical Engineering, Mechanical Engineering, Brigham and Women's Hospital



Present

Future

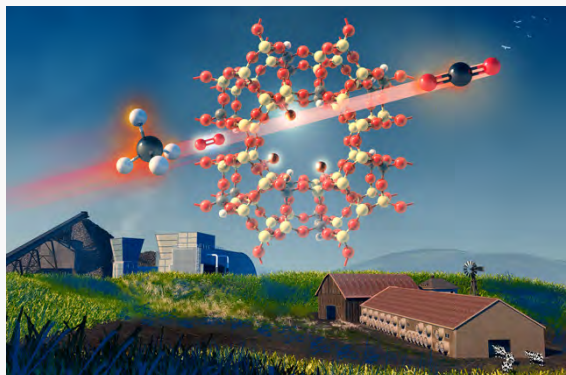
Links:

<https://doi.org/10.1126/sciadv.abm8478>

<https://doi.org/10.1016/j.fhfh.2022.100058>

<https://news.mit.edu/2022/gels-drugs-medicine-0527>

VAMCO: Removing Methane From Dairy Barns, Concentrated streams to Reduce the Rate of Climate Change

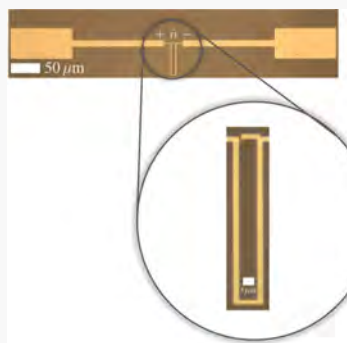


PIs: Desiree Plata, Tim Swager

Department: Civil and Environmental Engineering

<https://news.mit.edu/2022/dirt-cheap-solution-common-clay-materials-may-help-curb-methane-emissions>
<https://www.theengineer.co.uk/common-clay-could-curb-methane-emissions/>
<https://www.fastcompany.com/90711976/how-an-ingredient-found-in-cat-litter-could-help-fight-climate-change>
<https://www.optimistdaily.com/2022/01/key-ingredient-in-cat-litter-could-help-us-reduce-methane-emissions/>

Quantum Computing Qubit: Taking fluxonium to the limit



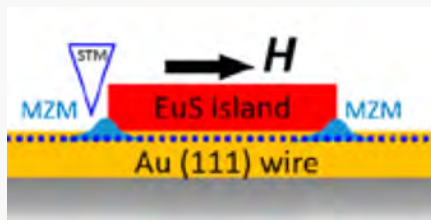
PIs: Neill Warrington Stephen Sorokanich

Departments: Physics. NIST

Links:

<https://journals.aps.org/prb/abstract/10.1103/PhysRevB.110.125404>
<https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.130.267001>

Majorana Qubits: Towards Fault-Tolerant for Quantum Information Processing



PI: Jegadeesh Moodera

Department: Physics

Links:

<https://doi.10.1073/pnas.1919753117>
<http://news.mit.edu/2020/first-majorana-fermion-metal-quantum-computing-0410>

OCAMLLM: Making LLMs Good in Math

	OccamLLM 8B / 70B	Llama 2 7B Chat	Llama 3 8b Instruct	GPT 3.5 Turbo	GPT 4o	GPT 4o Code
Addition	100.0 ± 0.0	19.2 ± 1.2	44.9 ± 1.6	65.2 ± 1.3	95.7 ± 0.6	100.0 ± 0.0
Subtraction	100.0 ± 0.0	8.7 ± 0.9	34.4 ± 1.8	59.8 ± 1.6	85.6 ± 1.1	99.5 ± 0.5
Multiplication	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	99.0 ± 0.7
Division	100.0 ± 0.0	2.8 ± 0.5	35.3 ± 1.5	10.7 ± 1.0	38.6 ± 1.5	100.0 ± 0.0
Square Root	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.9 ± 0.1	18.6 ± 1.2	100.0 ± 0.0
Exponential	100.0 ± 0.0	0.3 ± 0.2	3.1 ± 0.5	12.5 ± 1.0	23.2 ± 1.3	100.0 ± 0.0
Logarithm	100.0 ± 0.0	0.1 ± 0.1	0.0 ± 0.0	17.1 ± 1.2	21.3 ± 1.3	100.0 ± 0.0
Sine	100.0 ± 0.0	7.6 ± 0.8	7.0 ± 0.8	13.4 ± 1.1	39.3 ± 1.5	100.0 ± 0.0
Cosine	100.0 ± 0.0	0.8 ± 0.3	1.5 ± 0.4	6.7 ± 0.8	32.8 ± 1.5	100.0 ± 0.0
AVERAGE	100.0 ± 0.0	4.4 ± 0.2	14.0 ± 0.8	20.7 ± 0.4	39.5 ± 0.5	99.8 ± 0.1

PI: Marin Soljacic

Department: Physics

Links:

<https://arxiv.org/abs/2406.06576v4>
<https://doi.org/10.48550/arXiv.2402.00157>

Scintillation Technology To Increase Imaging Resolution And Efficiency



PI: Marin Soljacic

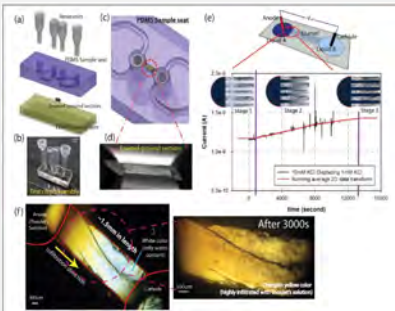
Department: Physics

Links:

<https://tlo.mit.edu/node/3812/pdf>
<https://www.sciencedirect.com/science/article/abs/pii/S0168900211001999#:~:text=The%20pixelated%20scintillation%20screen%20as,resolution%20of%20X%2Dray%20imaging.>

Electrical Engineering & Computer Science

Enhanced Transport of Materials Into Enamel Nanopores



PI: Jongyoon Han (J), Micro-Nanofluidic BioMEMS group

Department:
Electrical
Engineering &
Computer Science;
Biological
Engineering

Links:

<https://doi.org/10.1016/j.dental.2019.08.118>

<https://doi.org/10.1177/0022034518818463>

Printed Solar Cells for Next Generation Photovoltaics

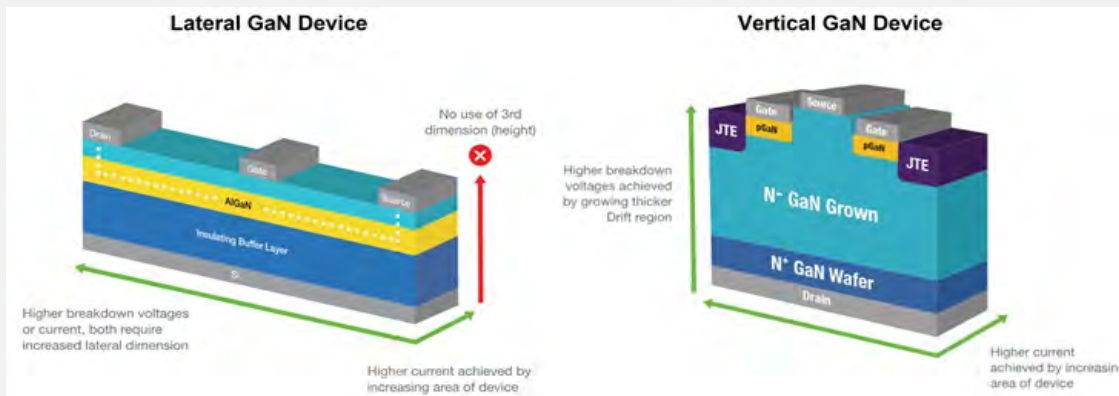


PI: Vladimir Bulovic

Department:
Electrical
Engineering &
Computer Science

<https://doi.org/10.1002/eem2.12043>

Vertical Gallium Devices for Future High-Power Electronics

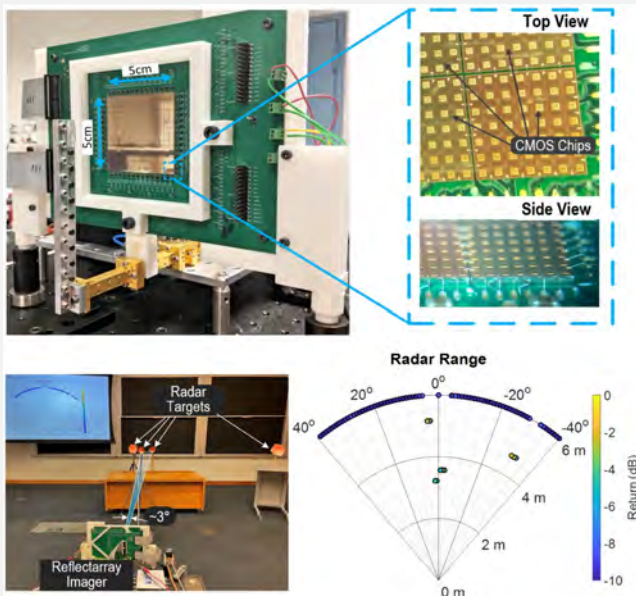


PI: Tomas Palacios

Department: Electrical
Engineering &
Computer Science

<https://www.eetimes.com/podcasts/vertical-gan/>

Terahertz Beam Steering Antenna Arrays



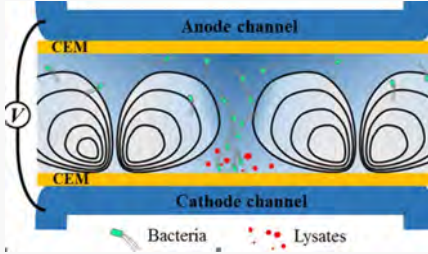
PI: Ruonan Han

Department: Electrical
Engineering & Computer
Science; Micro Technologies
Laboratory

Link:

<https://link.springer.com/article/10.1007/s10762-022-00902-1>

Electrochemical Lysis of Bacterial Pathogens



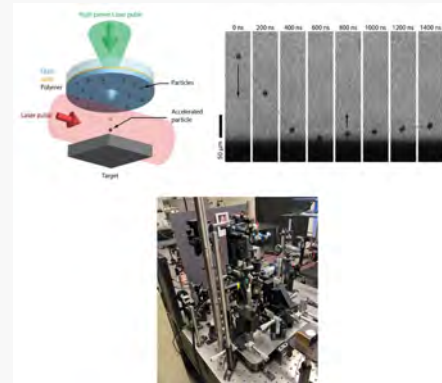
PI: Jongyoon Han (J)

Departments:
Electrical
Engineering and
Computer Science/
Biological
Engineering

Links:

<https://tlo.mit.edu/technologies/electromechanical-lysis-bacterial-pathogens-using-ion-concentration-polarization>

Laser-Induced Particle Impact Test ("LIPIT")



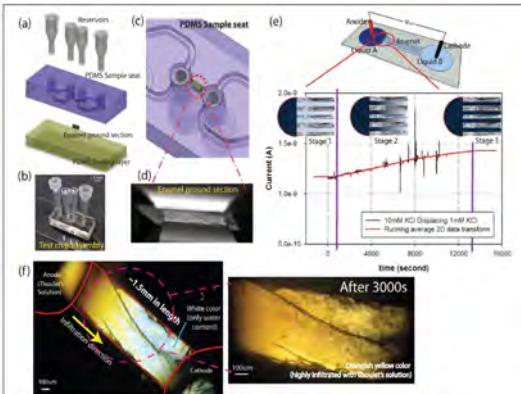
PIs: Christopher
Schuh
Keith Nelson,
Jasper Leinhard,
Yuchen Sun

Departments:
DMSE/Chemistry

Links

<https://news.mit.edu/2018/microscopic-materials-take-tiny-hits-1129>

Enhanced Transport of Materials Into Enamel Nanopores Via Electrokinetic Flow

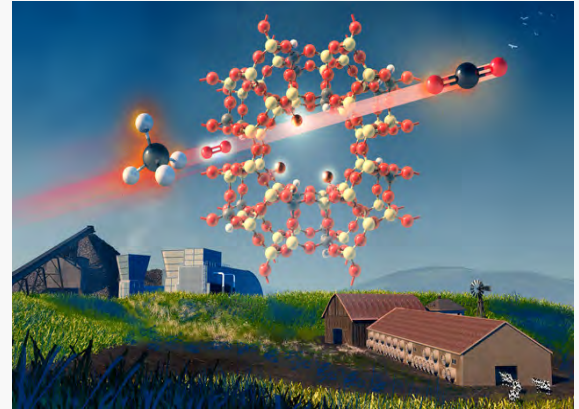


PI: Jongyoon Han (J)
Departments: EECS/BioE
Links:

<https://doi.org/10.1016/j.dental.2019.08.118>

<https://doi.org/10.1177/0022034518818463>

Removing Methane from Dairy Barns/ VAMCO: Concentrated Streams to Reduce the Rate of Climate Change



PIs:Desiree Plata, Tim Swager

Departments:
Civil and Environmental Engineering/ Chemistry

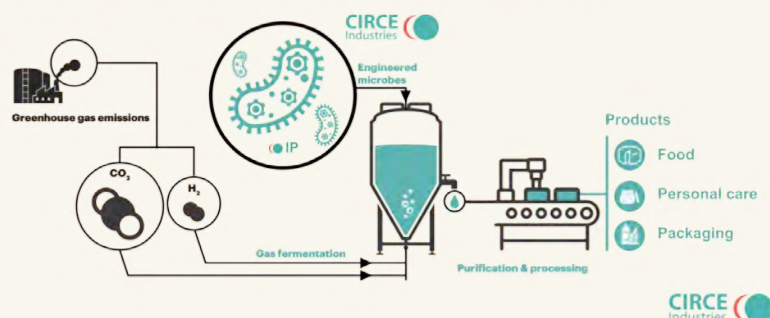
Links:

<https://news.mit.edu/2022/dirt-cheap-solution-common-clay-materials-may-help-curb-methane-emissions>

<https://www.theengineer.co.uk/common-clay-could-curb-methane-emissions/>

CIRCE Industries (Genetically Engineered Microbes Manufacturing Platform)

Circe makes bioproduction resilient, sustainable, and scalable



PI: Shannon Nangle

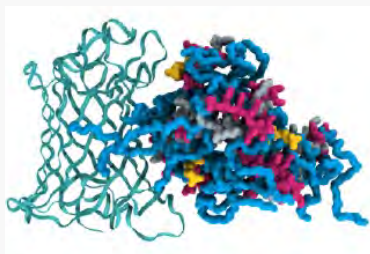
Department: Harvard Wyss Institute

Link:

<https://phys.org/news/2020-04-microbes-biodegradable-plastics-environmental-impact.html>

Materials Science and Engineering

Active Polymer Additives



PI: Alfredo
Alexander-Katz

Department: Material
Sciences and
Engineering

Links:

<https://chemrxiv.org/engage/chemrxiv/article-details/671123ff51558a15ef59f01a>
<https://www.sciencedirect.com/science/article/pii/S0032386122009910>
<https://pubs.acs.org/doi/full/10.1021/acs.macromol.1c02411>

Sharpest Blades



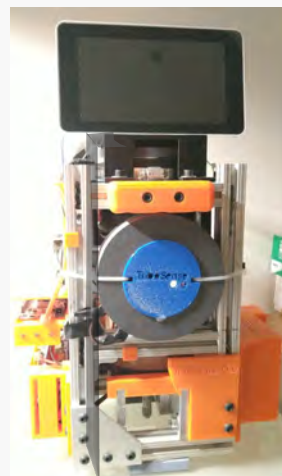
PI: Cem Tasan

Department: Material
Sciences and
Engineering

Links:

<https://news.mit.edu/2020/why-shaving-dulls-razors-0806>
<https://www.science.org/doi/10.1126/science.aba9490>

Biomolecular Sensing Platform for Precision Medicine

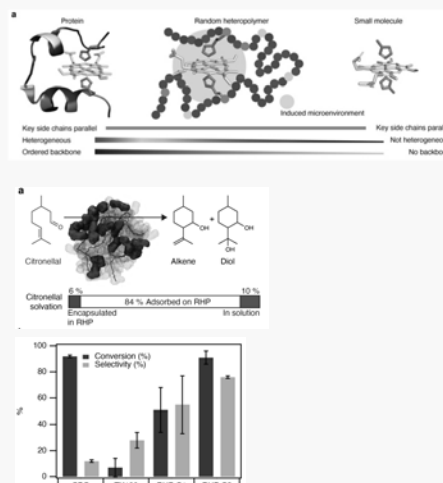


PI: Alfredo Alexander-Katz

Department: Material
Sciences and
Engineering

<https://www.biorxiv.org/content/10.1101/2021.02.05.429999v1.full>

Biomimetic Polymers as Synthetic Enzymes



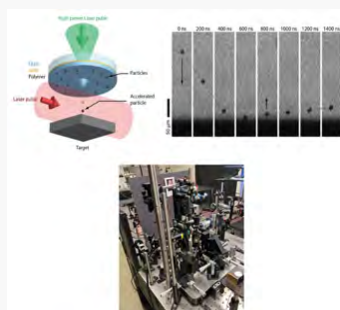
PI: Alfredo
Alexander-Katz

Department:
Material Sciences
and Engineering

Links:

<https://pubs.acs.org/doi/full/10.1021/acs.macromol.2c02413>
<https://www.nature.com/articles/s41586-022-05675-0>

Laser- Induced Particle Impact Test ("LIPIT")



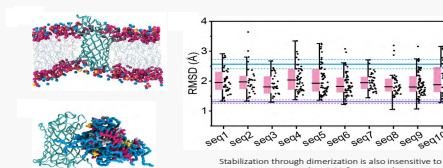
PIs: Chris Schuh, Keith Nelson

Department: Material
Sciences and
Engineering; Chemistry

Link:

<https://news.mit.edu/2018/microscopic-materials-take-tiny-hits-1129>
<https://news.mit.edu/2017/how-get-sprayed-metal-coatings-stick-1122>

Polymer Protectants for Unleashing Enzymes in Many Environments.



PI: Alfredo
Alexander-Katz

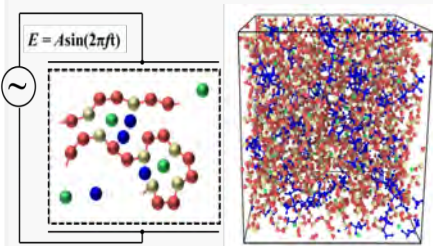
Department:
Material Sciences
and Engineering

Link:

<https://www.sciencedirect.com/science/article/abs/pii/S0032386122009910>

Materials Science and Engineering

Active Polymer Electrolytes for Batteries



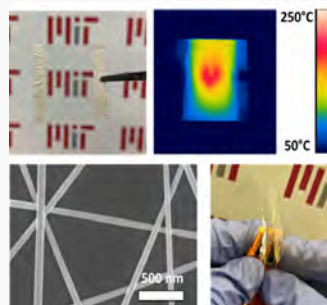
PI: Alfredo Alexander-Katz

Department: Material Sciences and Engineering

Link:

<https://www.sciencedirect.com/science/article/pii/S1364032123009942>

Oxynitride-Based Encapsulation of Silver Nanowire Networks

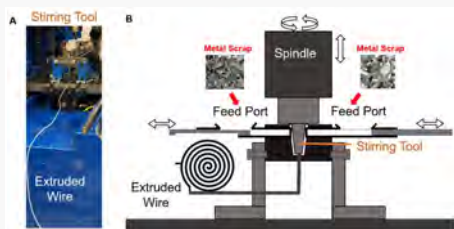


PI: Jeffery Grossman

Department: Material Sciences and Engineering

<https://pubmed.ncbi.nlm.nih.gov/35029366/>
<https://pubmed.ncbi.nlm.nih.gov/32233415/>

Revolutionizing Titanium: A Green Recycling Solution for Climate-Resilient Supply Chains



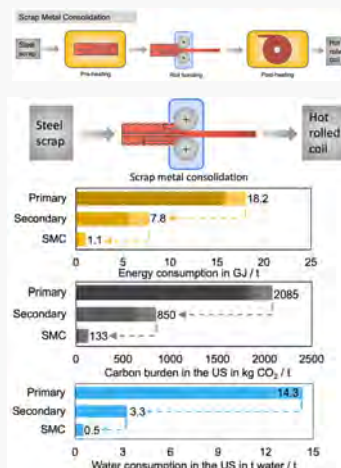
PIs : Chris Schuh
Hang Yu

Department: Material Sciences and Engineering

Link:

<https://www.sciencedirect.com/science/article/pii/S2772369022000755>

Recycling Steel: Scrap Metal Consolidation



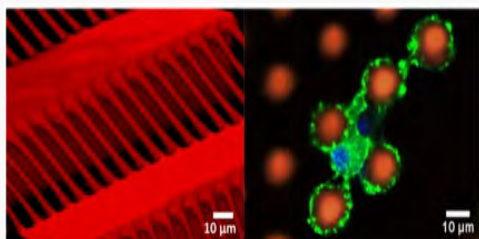
PI: Cem Tasan

Department: Material Sciences and Engineering

Link:

<https://news.mit.edu/2023/technologies-water-conservation-treatment-move-closer-commercialization-0906>

Artificial Axons



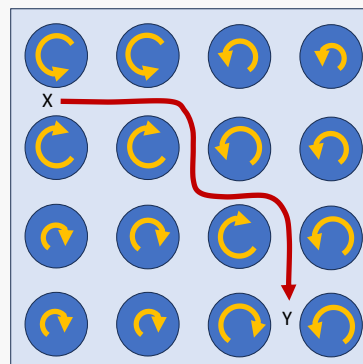
PIs: Anna Jagielska
Krystyn J. Van Vliet
Nicholas Fang

Department: Material Sciences and Engineering

Link:

<https://kjavgroup.mit.edu/wp-content/uploads/2020/08/Espinosa-Hoyos-et-al.-2018-Paper.pdf>
<https://news.mit.edu/2021/3-questions-anna-jagielska-mimicking-axons-0629>
<https://deshpande.mit.edu/projects/artificial-axons-as-a-myelination-assay-for-drug-screening-in-neurological-diseases/>

Fluidic Chips



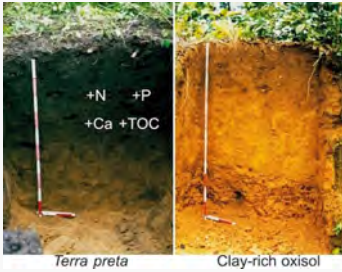
PI: Alfredo Alexander-Katz

Department: Material Sciences and Engineering

<https://arxiv.org/abs/2402.17707>
<https://pubs.rsc.org/en/content/articlehtml/2025/nr/d4nr02225d>

Problem Statements, IP Rescue, and Others

Living Off The Grid? Energy Independence? Amazon Rainforest?



PIs: Luis P Brevia, Doug Hart

Department: iTeams, Chemical Engineering, Mechanical Engineering

Links:

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3408238/>
<https://news.mit.edu/2021/using-aluminum-and-water-to-make-clean-hydrogen-fuel-0812>
<https://meche.mit.edu/news-media/2013-engineering-systems-design>



Obfuscation

PI: Luis P Brevia

Department: iTeams

Links:

<https://www.nytimes.com/2020/09/08/opinion/international-baccalaureate-algorithm-grades.html>

Reference Title:

<https://mitpress.mit.edu/books/obfuscation>



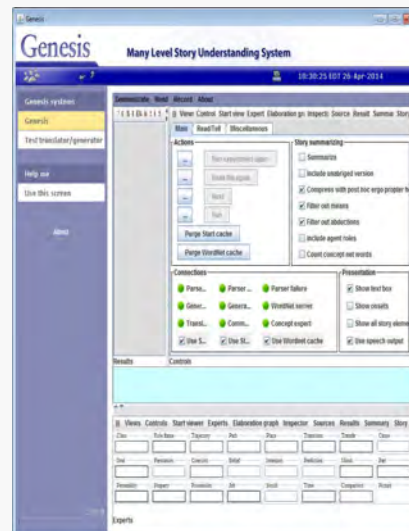
Genesis Project

PI: Luis P Brevia

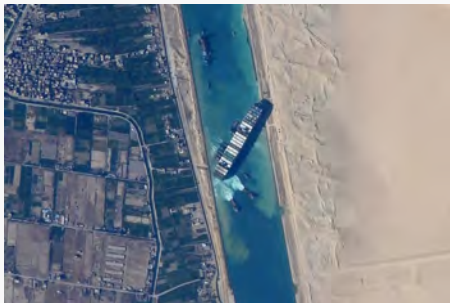
Dept: iTeams

Link:

<https://www.csail.mit.edu/research/genesis-group>
<https://www.csail.mit.edu/node/5891>
<https://groups.csail.mit.edu/genesis/papers.html>
<https://groups.csail.mit.edu/genesis/documentation.html>



Local Manufacturing



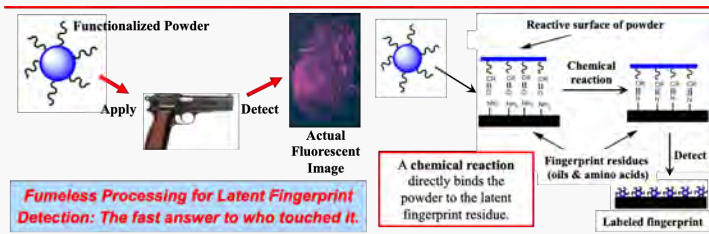
PI: Luis P Brevia

Department: iTeams

Link:

<https://www.bbc.com/news/world-us-canada-66321392>

Fumeless Latent Fingerprinting



PI: Lee Cambrea

Department: NAVAIR

Link:

<https://patents.google.com/patent/US8574658B1/en>



RAPID REACTION
Technology Office

Towards a Carbon-Capture Based Economy, A Carbon Currency



PI: Luis P Brevia

Department: iTeams
Chemical Engineering

Link:

<https://heinonline.org/HOL/LandingPage?handle=hein.journals/helr32&div=17&id=&page=>